FNCE30007 Mid-sem test A

Derivative Securities (University of Melbourne)

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Student Number:
Seat Number:

THE UNIVERSITY OF MELBOURNE
DEPARTMENT OF FINANCE

Mid-Semester Test—SEMESTER 1, 2024

SUBJECT NUMBER:	FNCE30007
SUBJECT NAME:	Derivative Securities
EXAMINATION DURATION:	1 hour
READING TIME:	5mins
TOTAL MARKS:	15
NO. OF PAGES:	9

(Do NOT write your name anywhere on this exam booklet)

AUTHORISED MATERIALS

1. This is a CLOSED BOOK examination.
2. Only Casio FX82 (any suffix) calculators are permitted. No other models of calculators will be permitted in the exam.
3. Students are NOT permitted to use any dictionary.

INSTRUCTIONS TO INVIGILATORS

1. Students should be provided ONLY with this booklet and an M1 Multiple Choice Answer Sheet.
2. Writing is NOT permitted during reading time.
3. This booklet must be COLLECTED at the end of the examination.

INSTRUCTIONS TO STUDENTS

1. Writing is NOT permitted during reading time.
2. Once writing is permitted, write your seat number and student number in the boxes at the top of this cover page.
3. Complete the M1 Multiple Choice Answer Sheet as follows:
   - Write your subject code and surname in the boxes provided on the sheet.
   - Write your Student Number under the words Student Number and then colour in the corresponding circles beneath the number. Do not put zeros at the beginning of the student ID on the multiple choice form. Begin with the first digit of your student ID.
4. Answer ALL questions. Record your answers on the M1 Multiple Choice Answer Sheet. Answers recorded in any other space will not be marked.
5. Pages titled "THIS PAGE HAS BEEN LEFT INTENTIONALLY BLANK" are provided for rough work. Examiners will NOT take into account anything written on these pages.
6. A formula list is provided on the last page. You may detach this page.
7. Do NOT detach any other pages.
8. **This paper must be handed in at the end of the examination. Do NOT remove this paper from the examination room.**

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1. Which of the following is true about a long forward contract
    A. The contract becomes more valuable as the price of the asset declines
    B. The contract becomes more valuable as the price of the asset rises
    C. The contract is worth zero if the price of the asset declines after the contract has been entered into
    D. The contract is worth zero if the price of the asset rises after the contract has been entered into
    E. None of the above

2. The price of a stock on February 1 is $124. A trader sells 200 put options on the stock with a strike price of $120 when the option price is $5. The options are exercised when the stock price is $110. The trader's net profit or loss is
    A. Gain of $1,000
    B. Loss of $2,000
    C. Loss of $2,800
    D. Loss of $1,000
    E. None of the above

3. A trader has a portfolio worth $5 million that mirrors the performance of a stock index. The stock index is currently 1,250. Futures contracts trade on the index with one contract being on $25 times the index. To remove market risk from the portfolio the trader should
    A. Buy 160 contracts
    B. Sell 160 contracts
    C. Buy 200 contracts
    D. Sell 200 contracts
    E. None of the above

4. You sell one December futures contracts when the futures price is $1,010 per unit. Each contract is on 100 units and the initial margin per contract that you provide is $2,000. The maintenance margin per contract is $1,500. During the next day the futures price rises to $1,012 per unit. What is the balance of your margin account at the end of the day?
    A. $1,800
    B. $3,300
    C. $2,200
    D. $3,700
    E. None of the above

5. When interest rates increase with all else remaining the same, which of the following is true?
    A. Both calls and puts increase in value
    B. Both calls and puts decrease in value

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C. Calls increase in value while puts decrease in value
D. Puts increase in value while calls decrease in value
E. None of the above

6. Which of the following is always true?
    A. An American call option on a stock should never be exercised early
    B. An American call option on a stock should never be exercised early when no dividends are expected
    C. There is always some chance that an American call option on a stock will be exercised early
    D. There is always some chance that an American call option on a stock will be exercised early when no dividends are expected
    E. None of the above

7. Interest rates are zero. A European call with a strike price of $50 and a maturity of one year is worth $6. A European put with a strike price of $50 and a maturity of one year is worth $7. The current stock price is $49. Which of the following is true?
    A. The call price is high relative to the put price
    B. The put price is high relative to the call price
    C. Both the call and put must be mispriced
    D. Both the call and put are priced correctly
    E. None of the above

8. The spot price of an investment asset is $30 and the risk-free rate for all maturities is 10% with continuous compounding. The asset provides an income of $2 at the end of the first year and at the end of the second year. What is the three-year forward price (to 2 decimal places)?
    A. $19.67
    B. $35.84
    C. $45.15
    D. $40.50
    E. None of the above

9. What should an arbitrageur do when the one-year forward price of an asset is too low? Assume that the asset provides no income.
    A. The arbitrageur should borrow the price of the asset, buy one unit of the asset and enter into a short forward contract to sell the asset in one year.
    B. The arbitrageur should borrow the price of the asset, buy one unit of the asset and enter into a long forward contract to buy the asset in one year.
    C. The arbitrageur should short the asset, invest the proceeds of the short sale at the risk-free rate, enter into a short forward contract to sell the asset in one year
    D. The arbitrageur should short the asset, invest the proceeds of the short sale at

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the risk-free rate, enter into a long forward contract to buy the asset in one year

E. None of the above

10. The basis is defined as spot minus futures. A trader is hedging the sale of an asset with a short futures position. The basis increases unexpectedly. Which of the following is true?
    A. The hedger's position improves.
    B. The hedger's position worsens.
    C. The hedger's position sometimes worsens and sometimes improves.
    D. The hedger's position stays the same.
    E. None of the above

11. On March 1 a commodity's spot price is $60 and its August futures price is $59. On July 1 the spot price is $64 and the August futures price is $63.50. A company entered into futures contracts on March 1 to hedge its purchase of the commodity on July 1. It closed out its position on July 1. What is the effective price (after taking account of hedging) paid by the company (to 2 decimal places)?
    A. $59.50
    B. $60.50
    C. $61.50
    D. $63.50
    E. None of the above

12. Which of the following increases basis risk?
    A. A large difference between the futures prices when the hedge is put in place and when it is closed out
    B. Dissimilarity between the underlying asset of the futures contract and the hedger's exposure
    C. A reduction in the time between the date when the futures contract is closed and its delivery month
    D. All of the above
    E. None of the above

13. A company will buy 1000 units of a certain commodity on June 30. It decides to hedge 80% of its exposure using futures contracts. The spot price is currently $100 and the August futures contract on the commodity is priced at $90. If the spot price and the futures price on June 30 turn out to be $112 and $110, respectively. What is the average price paid for the commodity?
    A. $92
    B. $96
    C. $102
    D. $106

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E. None of the above

14. Suppose that the risk-free rate is 10% per annum with continuous compounding and that the dividend yield on a stock index is 4% per annum. The index is standing at 400, and the futures price for a contract deliverable in four months is 405. What arbitrage opportunities, including positions to be taken in both the futures and spot markets, does this create? (to 2 decimal places)?
    A. Go long on the futures contract and short the stock underlying the index to lock in a profit of 3.08.
    B. Go short on the futures contract and long the stock underlying the index to lock in a profit of 3.08.
    C. Go long on the futures contract and long the stock underlying the index to lock in a profit of 3.08.
    D. Go short on the futures contract and short the stock underlying the index to lock in a profit of 3.08.

15. A company has a $36 million portfolio with a beta of 1.2. The futures price for a contract on an index is 9000. Futures contracts on $25 times the index can be traded. What trade is necessary to reduce beta to 0.9?
    A. Long 192 contracts
    B. Short 192 contracts
    C. Long 48 contracts
    D. Short 48 contracts
    E. None of the above

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FORMULAE SHEET

Cost of Asset/ Price Realised = $F_0 + (S_1 - F_1) = F_0 + \text{Basis}_1$

$F_0 + (S^*_1 - F_1) + (S_1 - S^*_1)$

$$N^* = \beta \frac{P}{A}$$

$$N^* = (\beta - \beta^*) \frac{P}{A}$$

$$r_c = m \cdot ln\left(1 + \frac{r_m}{m}\right)$$

$$F_0 = S_0 e^{rT}$$

$$F_0 = (S_0 + Q)e^{rT}$$

$$F_0 = S_0 e^{(r+q)T}$$

$$F_0 = (S_0 - D)e^{rT}$$

$$F_0 = S_0 e^{(r-d)T}$$

$$F_0 = S_0 e^{(r-r_f)T}$$

$$F_0 = S_0 e^{(r+q-y)T}$$

$$f = (F - K)e^{-rT}$$

$$f = (K - F)e^{-rT}$$

$c \leq S_0$

$C \leq S_0$

$p \leq Ke^{-rT}$

$P \leq K$

$c \geq max(S_0 - Ke^{-rT}, 0)$

$p \geq max(Ke^{-rT} - S_0, 0)$

$$c + Ke^{-rT} = p + S_0$$

$$c \geq \max(S_0 - D - Ke^{-rT}, 0)$$

$$p \geq \max(D + Ke^{-rT} - S_0, 0)$$

$S_0 - K < C - P < S_0 - Ke^{-rT}$

$c + D + Ke^{-rT} = p + S_0$

$S_0 - D - K < C - P < S_0 - Ke^{-rT}$

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